

Optimization of Graph Squares

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Abstract

In this paper, we investigate the matrix properties of graph squares, focusing in particular on the relationship between the adjacency matrix of a graph and the corresponding adjacency of its square.

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1 Introduction

The problem of finding the square root of a graph is a well-studied field in both mathematics and computer-science, with specific relation to distributed computing, network design and computational biology. Research in this area has primarily focused around the undirected graph H and its square $G = H^2$ which is obtained by adding an edge between every pair of vertices whose distance in H is at most two.

Much of the existing research on graph square roots has centered on characterizing when a graph admits such a root and on the complexity of reconstructing H from G . In contrast rather than studying the construction problem directly, we analyze the relationship between the adjacency

matrix A_H of the root and the adjacency matrix A_G of the square, and show that the problem can be completely categorized as an integer program.

We then examine the

2 Background

Let $H = (V, E_H)$ be the undirected *graph root* with $|V| = n$ vertices, and let $G = (V, E_G)$ be the undirected *graph square*. The *square* of the root graph H is denoted as $H^2 = (V, E_{H^2})$ with edge set

$$E_{H^2} = \{uv \mid 1 \leq d_H(u, v) \leq 2\}, \quad (1)$$

where $d_H(u, v)$ is the *distance metric* on graph H denoting the number of edges in the shortest path from vertex u to v . That is, edge $ij \in E_{H^2}$ exists if and only if $i \neq j$ and there is a path in H of length 1 or 2 from i to j .

Let $\mathbb{1}_{\geq 1} : \mathbb{R}^{n \times n} \rightarrow \{0, 1\}^{n \times n}$ denote the entrywise *indicator function*, defined by

$$\mathbb{1}_{\geq 1}(M)_{ij} = \begin{cases} 1 & \text{if } M_{ij} \geq 1, \\ 0 & \text{if } M_{ij} = 0. \end{cases}$$

Proposition 1. $G = H^2$ such the adjacency matrix of the graph square

$$G : A_G = \mathbb{1}_{\geq 1}((A_H)^2 - D_H + A_H). \quad (2)$$

is equivalent to (1)

$$H^2 : ij \in E_{H^2} \iff 1 \leq d_H(i, j) \leq 2. \quad (3)$$

Proof. We first show that any feasible root H that satisfies (3) must must also satisfy (2).

Let $ij \in E_{H^2}$ such that there is a path in H of length one or two from i to j . If there is a path of length one in H , then $(A_H)_{ij} = 1$. Furthermore, since the degree matrix is diagonal $(D_H)_{ij} = 0$, and as $(A_H)_{ij}^2 \geq 0$ as a square, $((A_H)^2 - D_H + A_H)_{ij} \geq 1$ such that $A_G = 1$.

We next show that any feasible root H that satisfies (2) must must also satisfy (3). □

We can convert the edge set E_{H^2} into the *adjacency matrix* of H^2 by setting

$$(A_{H^2})_{ij} = \begin{cases} 1, & \text{if } ij \in E_{H^2}, \\ 0, & \text{otherwise.} \end{cases} \quad (4)$$

This definition follows strictly from $(A_H^2)_{ij}$ giving the number of paths of length two between vertices i, j and with the *degree matrix* $(D_H)_{ii} = \deg_H(i)$ such that

$$A_{G=H^2} = \frac{(A_H)^2 - D_H}{\# \text{ paths between } i, j} + A_H.$$

2.1 Integer Program

Let $x_{ij} \in \{0, 1\}$ be the decision variables of A_H , let $y_{ikj} \in \{0, 1\}$ be the decision variable for $x_{ik}x_{kj}$ in $(A_H)_{ij}^2$ and let $z_{ij} \in \mathbb{Z}_{\geq 0}$ represent the argument inside the indicator function.

$$D_H = \begin{bmatrix} \deg_H(1) & 0 & \cdots & 0 & 0 \\ 0 & \ddots & \vdots & \ddots & \vdots \\ \vdots & \cdots & \deg_H(i) & \cdots & 0 \\ 0 & \ddots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & \cdots & \deg_H(n) \end{bmatrix}, A_H = \begin{bmatrix} x_{11} & \cdots & x_{1j} & \cdots & x_{1n} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ x_{i1} & \cdots & x_{ij} & \cdots & x_{in} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ x_{n1} & \cdots & x_{nj} & \cdots & x_{nn} \end{bmatrix} \text{ such that}$$

$$(A_H)^2 = \begin{bmatrix} \sum_{k=1}^n x_{1k}x_{k1} & \cdots & \sum_{k=1}^n x_{1k}x_{kj} & \cdots & \sum_{k=1}^n x_{1k}x_{kn} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \sum_{k=1}^n x_{ik}x_{k1} & \cdots & \sum_{k=1}^n x_{ik}x_{kj} & \cdots & \sum_{k=1}^n x_{ik}x_{kn} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \sum_{k=1}^n x_{nk}x_{k1} & \cdots & \sum_{k=1}^n x_{nk}x_{kj} & \cdots & \sum_{k=1}^n x_{nk}x_{kn} \end{bmatrix}.$$

Then with known constants $g_{ij} \in \{0, 1\}$ from A_G , can reformulating (2) as an integer program by equating entries of each matrix to give the following linear system of n^2 equations.

$$\begin{bmatrix} g_{11} \\ \vdots \\ g_{1j} \\ \vdots \\ g_{1n} \\ \vdots \\ g_{ij} \\ \vdots \\ g_{nn} \end{bmatrix} = \begin{bmatrix} \mathbb{1}_{\geq 1}(z_{11}) \\ \vdots \\ \mathbb{1}_{\geq 1}(z_{1j}) \\ \vdots \\ \mathbb{1}_{\geq 1}(z_{1n}) \\ \vdots \\ \mathbb{1}_{\geq 1}(z_{ij}) \\ \vdots \\ \mathbb{1}_{\geq 1}(z_{nn}) \end{bmatrix} = \begin{bmatrix} \mathbb{1}_{\geq 1}\left(\sum_{k=1}^n x_{1k}x_{k1} - (D_H)_{11} + x_{11}\right) \\ \vdots \\ \mathbb{1}_{\geq 1}\left(\sum_{k=1}^n x_{1k}x_{kj} - (D_H)_{1j} + x_{1j}\right) \\ \vdots \\ \mathbb{1}_{\geq 1}\left(\sum_{k=1}^n x_{1k}x_{kn} - (D_H)_{1n} + x_{1n}\right) \\ \vdots \\ \mathbb{1}_{\geq 1}\left(\sum_{k=1}^n x_{ik}x_{kj} - (D_H)_{ij} + x_{ij}\right) \\ \vdots \\ \mathbb{1}_{\geq 1}\left(\sum_{k=1}^n x_{nk}x_{kn} - (D_H)_{nn} + x_{nn}\right) \end{bmatrix}. \quad (5)$$

The matrix D_H is diagonal can be represented as $(D_H)_{ii} = \sum_k x_{ik}$ where $(D_H)_{ij} = 0$ for $i \neq j$. Let $M \in \mathbb{Z}_{>0}$ be a sufficiently large constant. The known z_{ij} represent the terms inside the indicator function such that

$$g_{ij} = 1 \iff z_{ij} \geq 1 \text{ and } g_{ij} = 0 \iff z_{ij} = 0.$$

The above condition can be modelled using a standard big- M formulation, and the resulting integer

program is given by

Variables:

$$x_{ij} \in \mathbb{Z}_{\geq 0}, \quad \forall i, j \in \{1, \dots, n\}, \quad (6)$$

$$y_{ikj} \in \mathbb{Z}_{\geq 0}, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (7)$$

$$z_{ij} \in \mathbb{Z}_{\geq 0}, \quad \forall i, j \in \{1, \dots, n\}. \quad (8)$$

Constraints:

$$(1) \text{ Indicator argument: } z_{ii} = \sum_{k=1}^n y_{iki} - \sum_{k=1}^n x_{ik} + x_{ii}, \forall i \in \{1, \dots, n\}, \quad (9)$$

$$z_{ij} = \sum_{k=1}^n y_{ikj} + x_{ij}, \quad \forall i \neq j \in \{1, \dots, n\}, \quad (10)$$

$$(2) \text{ Indicator function: } z_{ij} \leq M g_{ij}, \quad \forall i, j \in \{1, \dots, n\}, \quad (11)$$

$$z_{ij} \geq g_{ij}, \quad \forall i, j \in \{1, \dots, n\}, \quad (12)$$

$$(3) \text{ Product: } y_{ikj} \leq x_{ik}, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (13)$$

$$y_{ikj} \leq x_{kj}, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (14)$$

$$y_{ikj} \geq x_{ik} + x_{kj} - 1, \quad \forall i, k, j \in \{1, \dots, n\} \quad (15)$$

$$(4) \text{ Upper Bound: } x_{ij} \leq 1, \quad \forall i, j \in \{1, \dots, n\}, \quad (16)$$

$$y_{ikj} \leq 1, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (17)$$

In addition, the adjacency matrix x must satisfy further constraints arising from symmetry and diagonality. Although analogous constraints could also be imposed on the auxiliary variables z they are omitted due to redundancy and potentially introduces artificial degeneracy in the linear program.

$$(5) \text{ Symmetry: } x_{ij} = x_{ji}, \quad \forall i, j \in \{1, \dots, n\}, \quad (18)$$

$$(6) \text{ Diagonal: } x_{ii} = 0, \quad \forall i \in \{1, \dots, n\}. \quad (19)$$

2.2 Linear Program

The linear relaxation of the integer program defined by Equations (6)–(19) is constructed in order to generate a feasible weighted solution in polynomial time algorithm. It is done with identical constraints, however the decision variables (6), (7) and (8) are replaced with

$$x_{ij} \in \mathbb{R}_{\geq 0}, \quad \forall i, j \in \{1, \dots, n\}, \quad (20)$$

$$y_{ikj} \in \mathbb{R}_{\geq 0}, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (21)$$

$$z_{ij} \in \mathbb{R}_{\geq 0}, \quad \forall i, j \in \{1, \dots, n\}. \quad (22)$$

The results from the linear programming can give *integer solutions* and *fractional solutions* when the root exists and *fractional solutions* and *infeasible solutions* when the root does not exist such that the results in Figure 1 can be generated.

However, the direct linear relaxation needs additional constraints after the linear relaxation.

Consider $z_{ii} = \sum_{k=1}^n y_{iki} - \sum_{k=1}^n x_{ik} + x_{ii}$ which can be simplified to $z_{ii} = \sum_{k=1}^n x_{ik} x_{ki} - \sum_{k=1}^n x_{ik}$.

Proposition 2.

3 Linear Program

3.1 Max Objective Function

Consider the addition of an objective function maximizing the total weights on all edges to the linear relaxation defined by Equations (9)–(22) in Section 2.2

$$\max \sum_{i=1}^n \sum_{j=1}^n x_{ij}. \quad (23)$$

This addition gives the improved results in Figure 2.

3.2 Singular Triangular Decomposition

Using the local triangular structure of the graph, certain edge assignments in a square root can be fixed. For a vertex $s \in V$, define

$$T(s) = \{\{u, v\} \subseteq V \setminus \{s\} \mid (s, u) \in E, (s, v) \in E, (u, v) \in E\}$$

to be the set of triangles in G that contain s . If $|T(s)| = 1$ where $T(s) = \{\{u, v\}\}$ then the vertex s exists in a unique triangle of G . When the neighbours are structurally distinct with different degree $\deg_G(u) \neq \deg_G(v)$, then the vertex with smaller degree must be the leaf-adjacent vertex in the square root. We enforce the deterministic assignment

$$x_{sw} = 1, x_{ws} = 1 \text{ where } w = \arg \min\{\deg_G(u), \deg_G(v)\}. \quad (24)$$

If $\deg_G(u) = \deg_G(v)$, no constraint is imposed. This addition gives the improved results in Figure 3 whilst maintaining polynomial runtime.

In the case of $\deg_G(u) = \deg_G(v)$, enforcement is done by arbitrarily choosing one of the edges of only one such triangle. This is done recursively until all such singular triangles are resolved. This gives the improved results in Figure 4.

Ross and Harary in 1960 gave an algorithm for constructing a tree root by using characterizations of the square graph to conclude that existing tree square roots are unique up to isomorphism [RH60].

Proposition 3. *Singular triangular decomposition completely categorizes all triangle roots in polynomial time linear program.*

3.3 Rounding

LP fractional solutions are derived from the LP when the root H exists and does not exist. An attempt for rounding is achieved such that the classification in Figure 5 is obtained.

3.4 Dual Problem

We now consider the dual formulation of the max objective linear program defined in Section 3.1 by first restating it in canonical form.

Objective Function:

$$\min - \sum_{i=1}^n \sum_{j=1}^n x_{ij}. \quad (25)$$

Variables:

$$x_{ij} \in \mathbb{R}_{\geq 0} \left([0, 1] \right), \quad \forall i, j \in \{1, \dots, n\}, \quad (26)$$

$$y_{ikj} \in \mathbb{R}_{\geq 0} \left([0, 1] \right), \quad \forall i, k, j \in \{1, \dots, n\}, \quad (27)$$

$$z_{ij} \in \mathbb{R}_{\geq 0}, \quad \forall i, j \in \{1, \dots, n\}. \quad (28)$$

Constraints:

$$(1) \text{ Indicator argument: } z_{ii} - \sum_{k=1}^n y_{iki} + \sum_{k=1}^n x_{ik} - x_{ii} = 0, \forall i \in \{1, \dots, n\}, \quad (29)$$

$$z_{ij} - \sum_{k=1}^n y_{ikj} - x_{ij} = 0, \quad \forall i \neq j \in \{1, \dots, n\}, \quad (30)$$

$$(2) \text{ Indicator function: } z_{ij} - Mg_{ij} \leq 0, \quad \forall i, j \in \{1, \dots, n\}, \quad (31)$$

$$-z_{ij} + g_{ij} \leq 0, \quad \forall i, j \in \{1, \dots, n\}, \quad (32)$$

$$(3) \text{ Product: } y_{ikj} - x_{ik} \leq 0, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (33)$$

$$y_{ikj} - x_{kj} \leq 0, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (34)$$

$$x_{ik} + x_{kj} - y_{ikj} - 1 \leq 0, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (35)$$

$$(4) \text{ Symmetry: } x_{ij} - x_{ji} = 0, \quad \forall i, j \in \{1, \dots, n\}, \quad (36)$$

$$(5) \text{ Diagonal: } x_{ii} = 0, \quad \forall i \in \{1, \dots, n\}, \quad (37)$$

$$(6) \text{ Variable Bound: } x_{ij} - 1 \leq 0, \quad \forall i, j \in \{1, \dots, n\}, \quad (38)$$

$$y_{ikj} - 1 \leq 0, \quad \forall i, k, j \in \{1, \dots, n\}. \quad (39)$$

This gives the dual problem with

Objective Function:

$$\max \sum_{i=1}^n \sum_{j=1}^n \left(-Mg_{ij}\gamma_{ij} + g_{ij}\delta_{ij} - \lambda_{ij} \right) - \sum_{i=1}^n \sum_{k=1}^n \sum_{j=1}^n (\eta_{ikj} + \mu_{ikj}). \quad (40)$$

Dual Variables:

$$\alpha_i \in \mathbb{R}, \quad \forall i \in \{1, \dots, n\}, \quad (\text{associated with (29)}), \quad (41)$$

$$\beta_{ij} \in \mathbb{R}, \quad \forall i \neq j \in \{1, \dots, n\}, (\text{associated with (30)}), \quad (42)$$

$$\gamma_{ij} \geq 0, \quad \forall i, j \in \{1, \dots, n\}, \quad (\text{associated with (31)}), \quad (43)$$

$$\delta_{ij} \geq 0, \quad \forall i, j \in \{1, \dots, n\}, \quad (\text{associated with (32)}), \quad (44)$$

$$\epsilon_{ikj} \geq 0, \quad \forall i, k, j \in \{1, \dots, n\}, (\text{associated with (33)}), \quad (45)$$

$$\zeta_{ikj} \geq 0, \quad \forall i, k, j \in \{1, \dots, n\}, (\text{associated with (34)}), \quad (46)$$

$$\eta_{ikj} \geq 0, \quad \forall i, k, j \in \{1, \dots, n\}, (\text{associated with (35)}), \quad (47)$$

$$\theta_{ij} \in \mathbb{R}, \quad \forall i, j \in \{1, \dots, n\}, \quad (\text{associated with (36)}), \quad (48)$$

$$\iota_i \in \mathbb{R}, \quad \forall i \in \{1, \dots, n\}, \quad (\text{associated with (37)}), \quad (49)$$

$$\lambda_{ij} \geq 0, \quad \forall i, j \in \{1, \dots, n\}, \quad (\text{associated with (38)}), \quad (50)$$

$$\mu_{ikj} \geq 0, \quad \forall i, k, j \in \{1, \dots, n\}, (\text{associated with (39)}). \quad (51)$$

Dual Constraints:

$\forall i, j \in \{1, \dots, n\}$, each primal x_{ij} :

$$\begin{cases} -1 - \sum_{k=1}^n \epsilon_{iik} - \sum_{k=1}^n \zeta_{kii} + \sum_{k=1}^n \eta_{iik} + \sum_{k=1}^n \eta_{kii} + \iota_i + \lambda_{ii} \geq 0, & \text{if } j = i, \\ -1 + \alpha_i - \beta_{ij} - \sum_{k=1}^n \epsilon_{ikj} - \sum_{k=1}^n \zeta_{kij} + \sum_{k=1}^n \eta_{ikj} + \sum_{k=1}^n \eta_{kij} + \theta_{ij} - \theta_{ji} + \lambda_{ij} \geq 0, & \text{if } j \neq i, \end{cases} \quad (52)$$

$\forall i, k, j \in \{1, \dots, n\}$, each primal y_{ikj} :

$$\begin{cases} -\alpha_i + \epsilon_{iki} + \zeta_{iki} - \eta_{iki} + \mu_{iki} \geq 0, & \text{if } j = i, \\ -\beta_{ij} + \epsilon_{ikj} + \zeta_{ikj} - \eta_{ikj} + \mu_{ikj} \geq 0, & \text{if } j \neq i, \end{cases} \quad (53)$$

$\forall i, j \in \{1, \dots, n\}$, each primal z_{ij} :

$$\begin{cases} \alpha_i + \gamma_{ii} - \delta_{ii} = 0, & \text{if } j = i, \\ \beta_{ij} + \gamma_{ij} - \delta_{ij} = 0, & \text{if } j \neq i. \end{cases} \quad (54)$$

whereby the Lagrangian is

$$\begin{aligned} \mathcal{L}(x, y, z) = & - \sum_{i=1}^n \sum_{j=1}^n x_{ij} + \sum_{i=1}^n \alpha_i \left(z_{ii} - \sum_{k=1}^n y_{iki} + \sum_{k=1}^n x_{ik} - x_{ii} \right) + \sum_{i \neq j} \beta_{ij} \left(z_{ij} - \sum_{k=1}^n y_{ikj} - x_{ij} \right) \\ & + \sum_{i=1}^n \sum_{j=1}^n \gamma_{ij} (z_{ij} - M g_{ij}) + \sum_{i=1}^n \sum_{j=1}^n \delta_{ij} (-z_{ij} + g_{ij}) + \sum_{i=1}^n \sum_{k=1}^n \sum_{j=1}^n \epsilon_{ikj} (y_{ikj} - x_{ik}) \\ & + \sum_{i=1}^n \sum_{k=1}^n \sum_{j=1}^n \zeta_{ikj} (y_{ikj} - x_{kj}) + \sum_{i=1}^n \sum_{k=1}^n \sum_{j=1}^n \eta_{ikj} (x_{ik} + x_{kj} - y_{ikj} - 1) \\ & + \sum_{i=1}^n \sum_{j=1}^n \theta_{ij} (x_{ij} - x_{ji}) + \sum_{i=1}^n \iota_i x_{ii} + \sum_{i=1}^n \sum_{j=1}^n \lambda_{ij} (x_{ij} - 1) + \sum_{i=1}^n \sum_{k=1}^n \sum_{j=1}^n \mu_{ikj} (y_{ikj} - 1). \end{aligned}$$

Collecting terms gives

$$\begin{aligned} \mathcal{L}(x, y, z) = & \sum_{i=1}^n \sum_{j=1}^n x_{ij} \left[-1 + \mathbf{1}_{\{j=i\}} \left(\alpha_i - \sum_{k=1}^n \epsilon_{iki} - \sum_{k=1}^n \zeta_{kii} + \sum_{k=1}^n \eta_{iki} + \sum_{k=1}^n \eta_{kii} + \iota_i + \lambda_{ii} \right) \right. \\ & \left. + \mathbf{1}_{\{j \neq i\}} \left(\alpha_i - \beta_{ij} - \sum_{k=1}^n \epsilon_{ikj} - \sum_{k=1}^n \zeta_{kij} + \sum_{k=1}^n \eta_{ikj} + \sum_{k=1}^n \eta_{kij} + \lambda_{ij} \right) + \theta_{ij} - \theta_{ji} \right] \\ & + \sum_{i=1}^n \sum_{k=1}^n \sum_{j=1}^n y_{ikj} \left[-\mathbf{1}_{\{j=i\}} \alpha_i - \mathbf{1}_{\{j \neq i\}} \beta_{ij} + \epsilon_{ikj} + \zeta_{ikj} - \eta_{ikj} + \mu_{ikj} \right] \\ & + \sum_{i=1}^n \sum_{j=1}^n z_{ij} \left[\mathbf{1}_{\{j=i\}} \alpha_i + \mathbf{1}_{\{j \neq i\}} \beta_{ij} + \gamma_{ij} - \delta_{ij} \right] \\ & - \sum_{i=1}^n \sum_{j=1}^n M g_{ij} \gamma_{ij} + \sum_{i=1}^n \sum_{j=1}^n g_{ij} \delta_{ij} - \sum_{i=1}^n \sum_{k=1}^n \sum_{j=1}^n \eta_{ikj} - \sum_{i=1}^n \sum_{j=1}^n \lambda_{ij} - \sum_{i=1}^n \sum_{k=1}^n \sum_{j=1}^n \mu_{ikj}. \end{aligned}$$

This dual formulation

A Result Diagrams

A.1 Linear Program

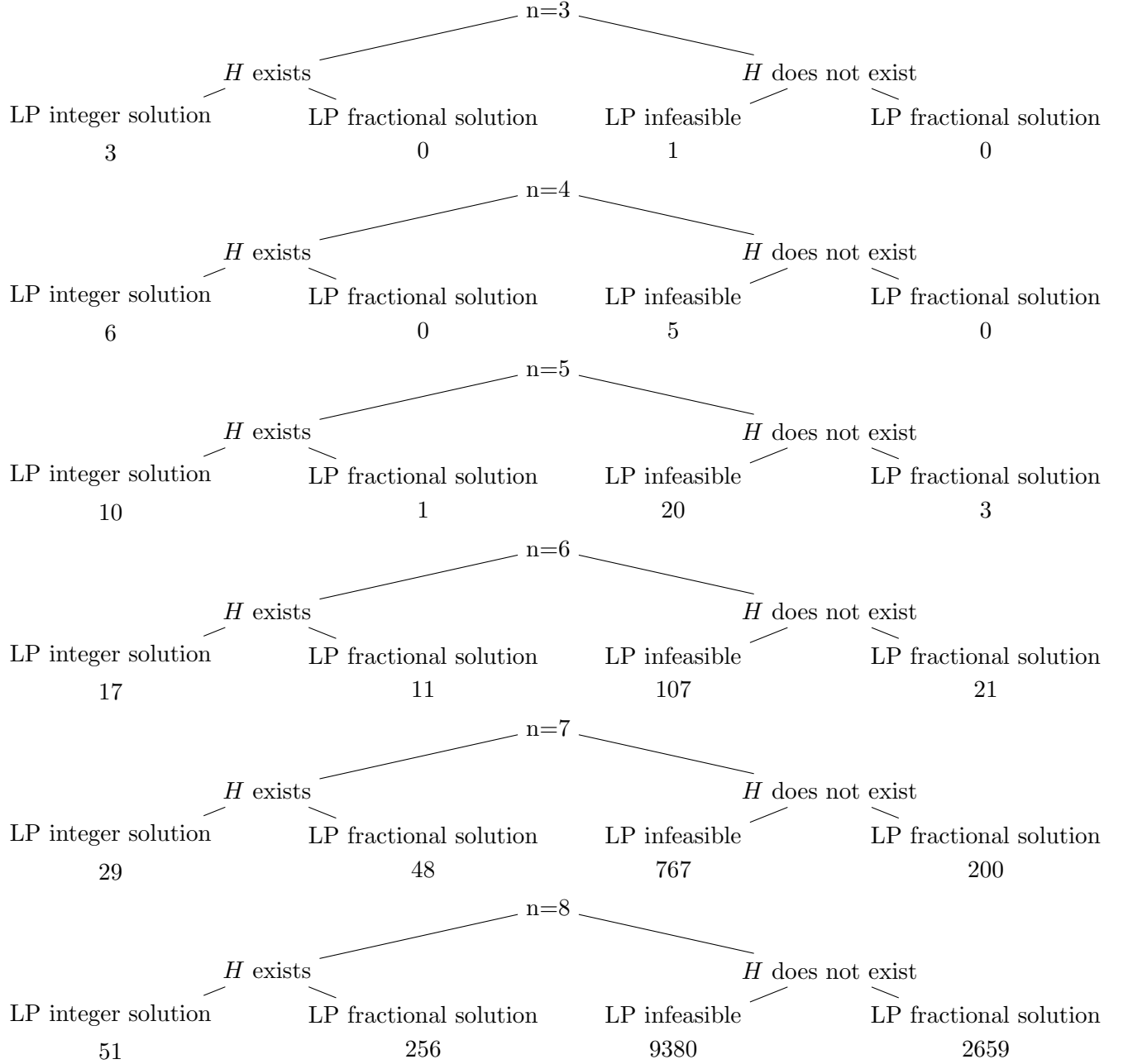


Figure 1: LP outcomes for different values of n on base linear program.

1

¹The total number of simple graphs on $n = 2, 3, 4, \dots$ unlabeled nodes is the sequence 2, 4, 11, 34, 156, 1044, 12346, $\dots$ given by A000088.

The IP integer solution sequence 2, 3, 6, 10, 17, 29, 51, $\dots$ is given by A285665.

The total number of graphs which are squares is the sequence 2, 3, 6, 11, 28, 77, 307, $\dots$ given by A382181.

A.1.1 Max Objective Function

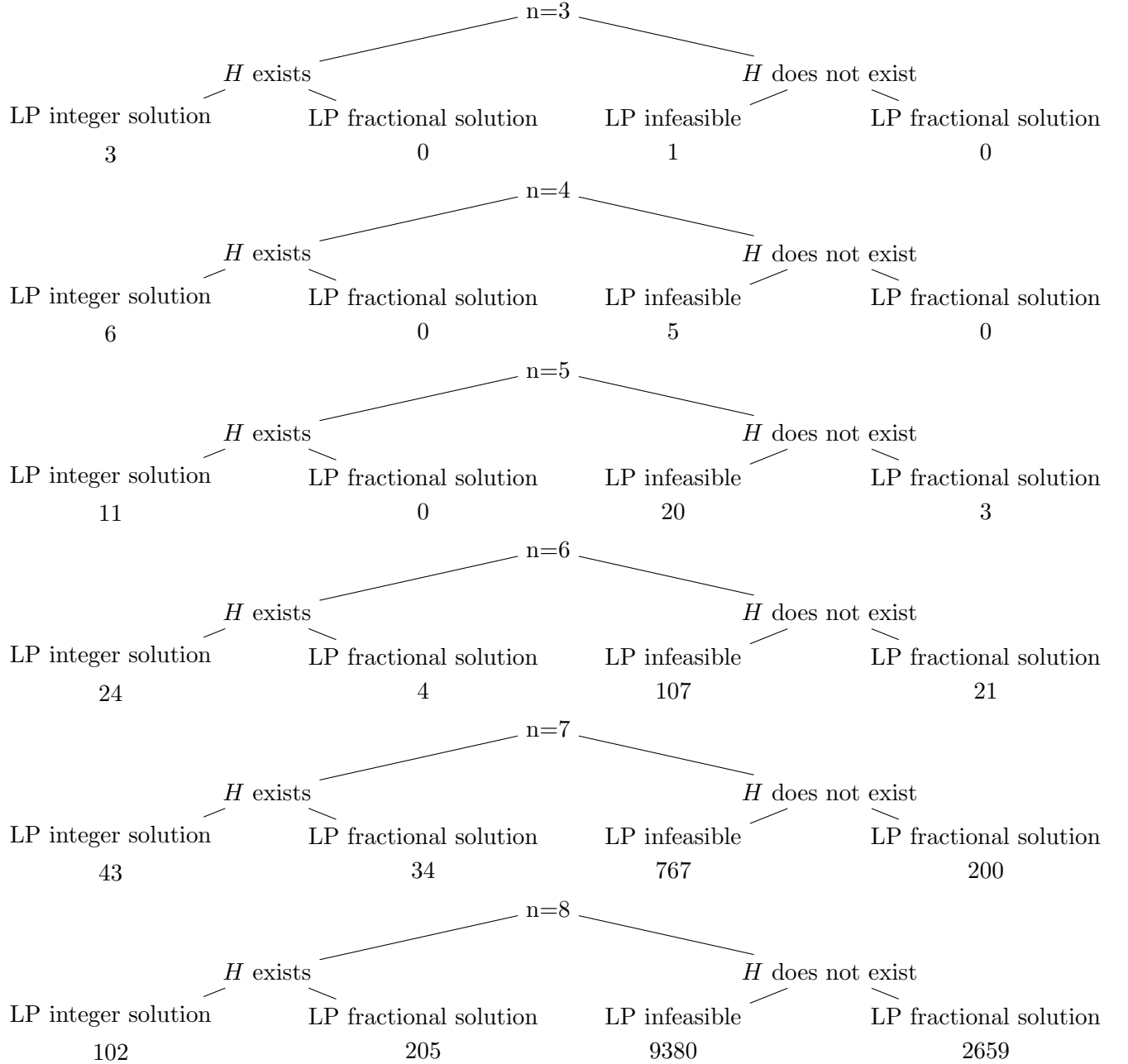


Figure 2: LP outcomes for different values of n on the linear program with max objective function.

2

²The LP integer solution sequence 3, 6, 11, 22, 42, 102, ... is very similar to the number of distinct characteristic polynomials of trees with n nodes - A226594.

A.1.2 Singular Triangular Decomposition

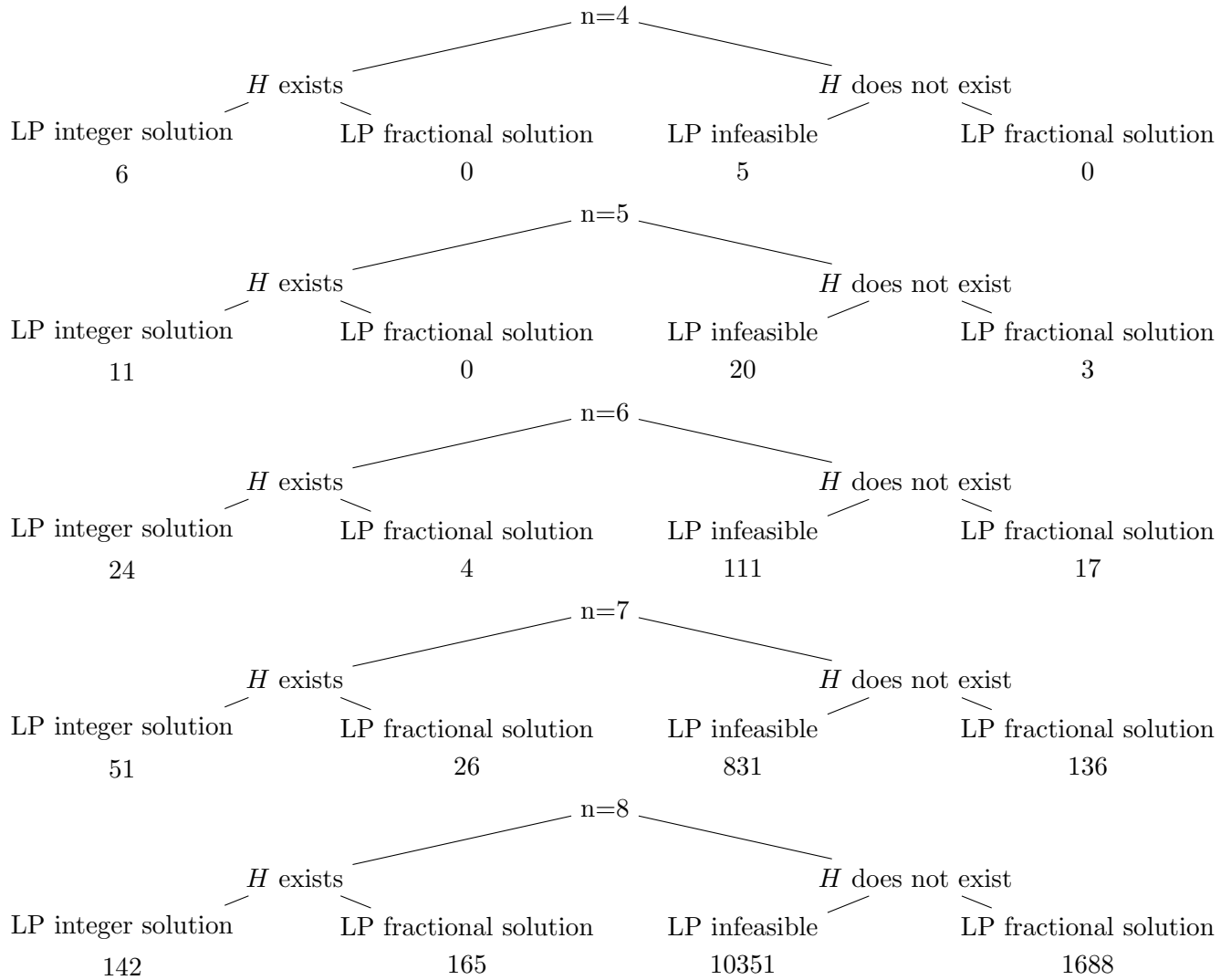


Figure 3: LP outcomes for different values of n on the linear program with max objective function and singular triangular decomposition.

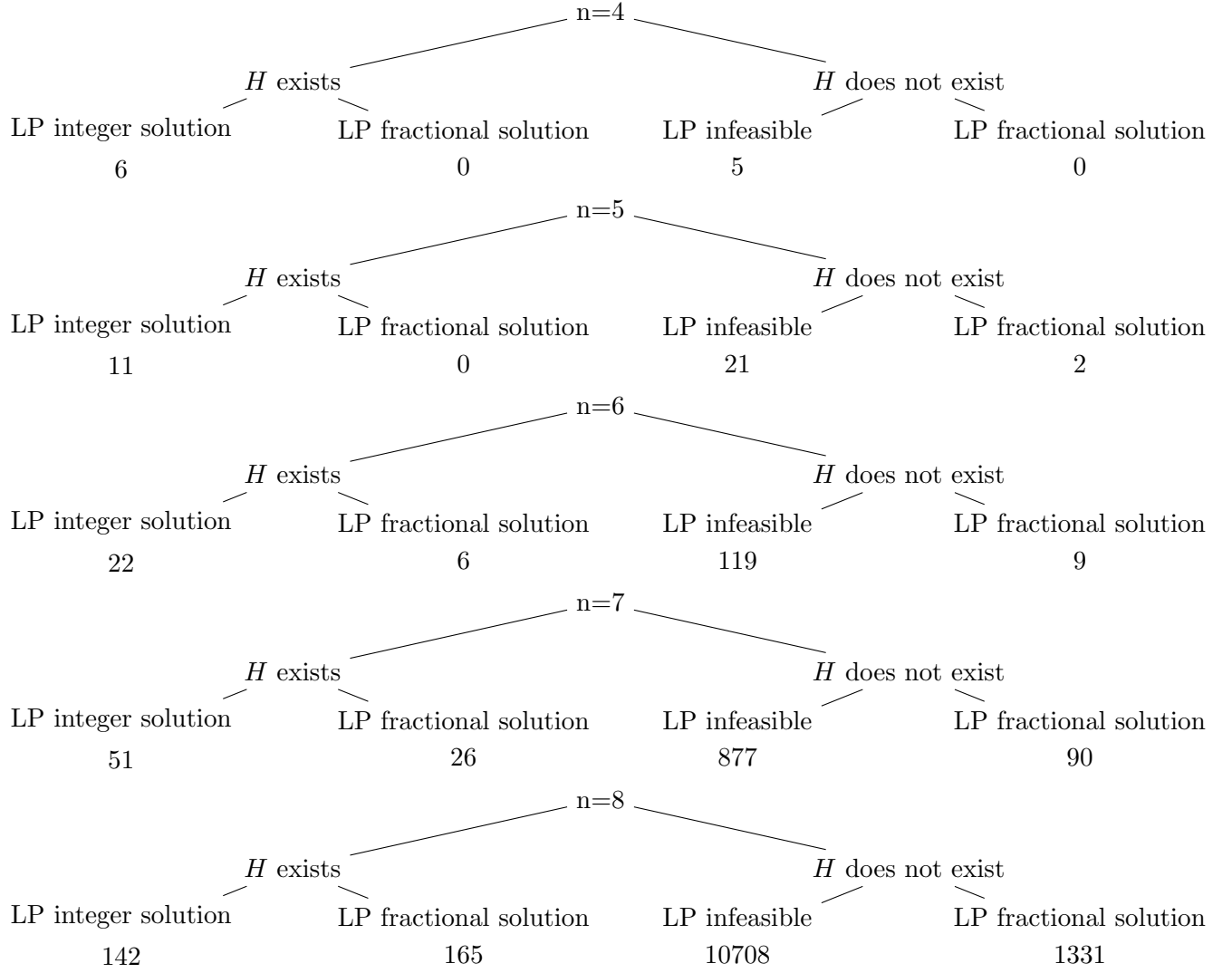


Figure 4: LP outcomes for different values of n on the linear program with max objective function and singular triangular decomposition with recursion for tie-breaking.

A.1.3 Rounding

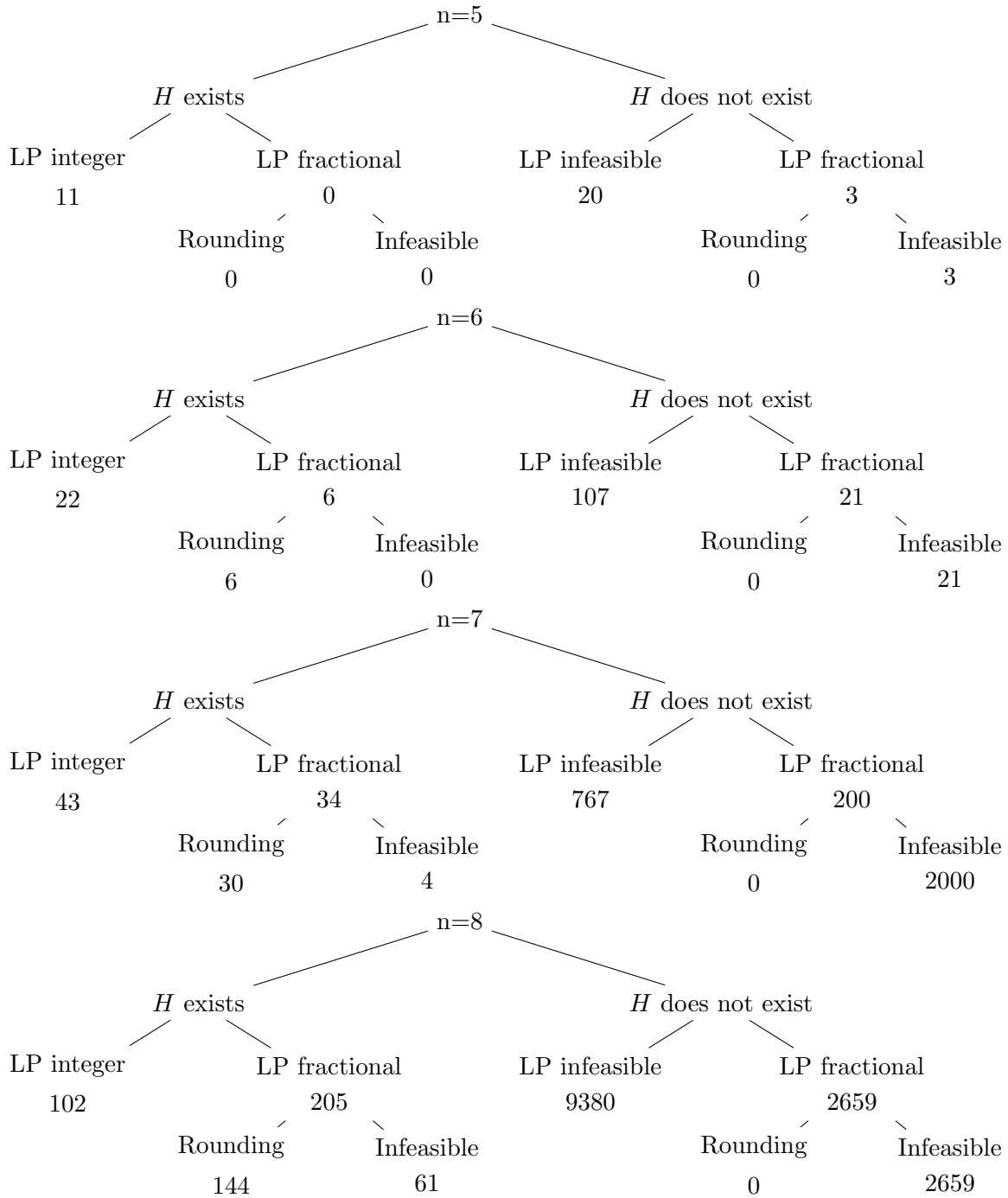


Figure 5: LP outcomes for different values of n on the linear program with max objective function and using rounding for fractional solutions.

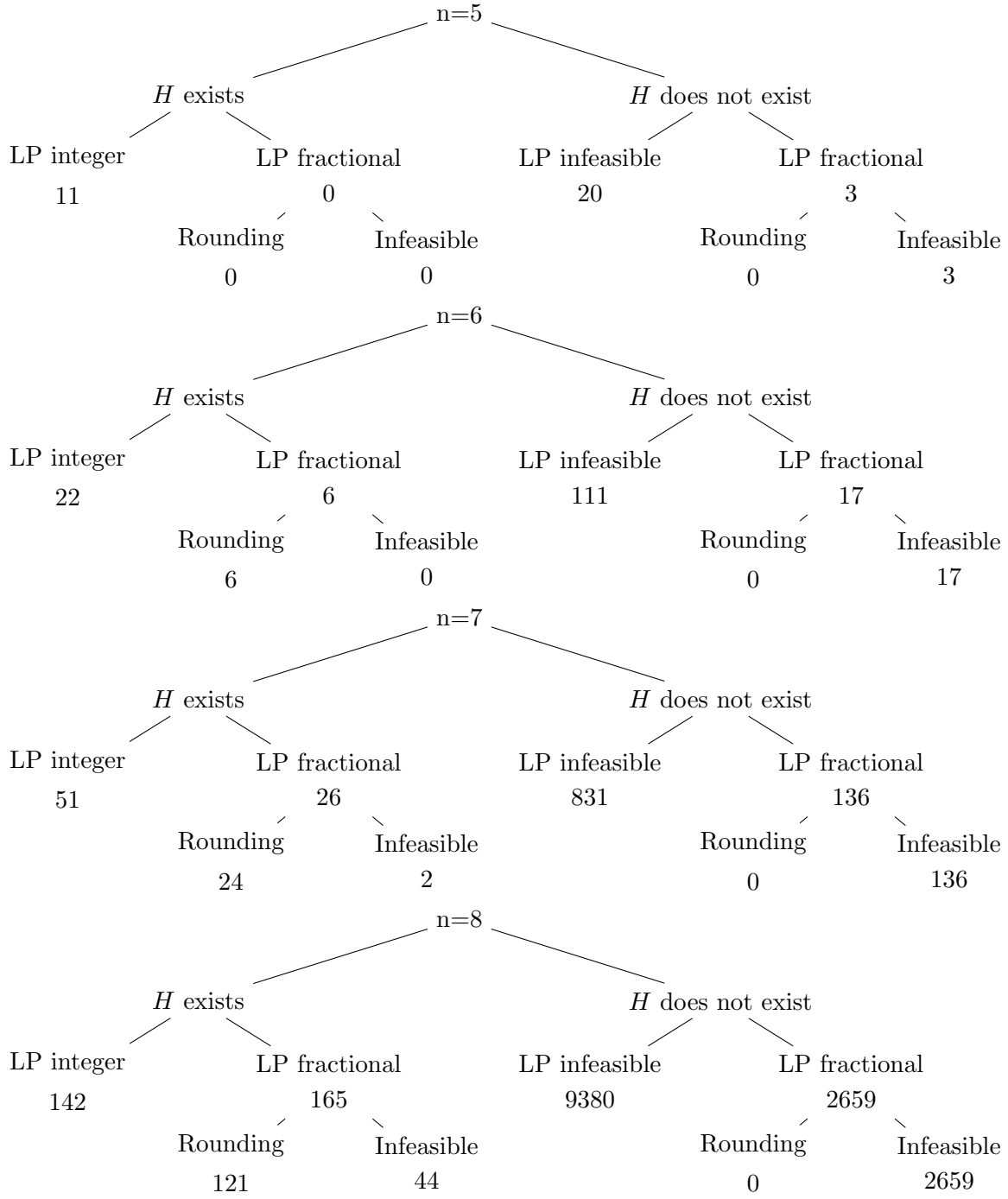


Figure 6: LP outcomes for different values of n on the linear program with max objective function, singular triangular decomposition and using rounding for fractional solutions.

A.2 Dual Program

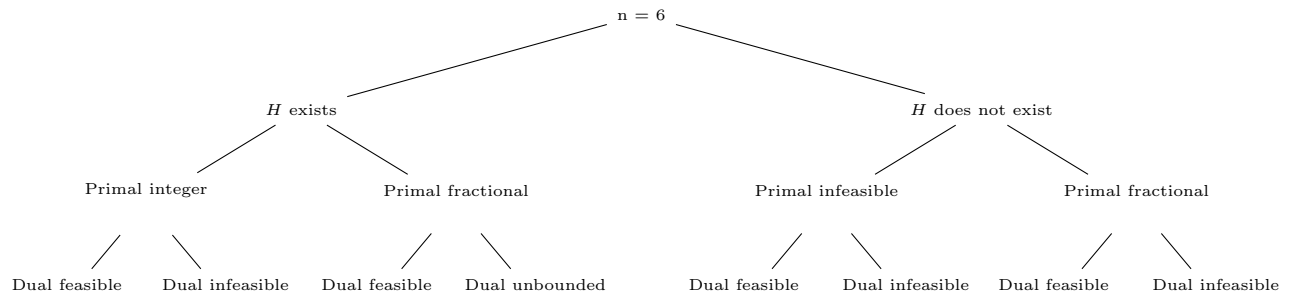


Figure 7: LP outcomes for different values of n on the dual linear program.

References

- [RH60] Ian C. Ross and Frank Harary. “The square of a tree”. In: *The Bell System Technical Journal* 39.3 (1960), pp. 641–647.